

Clement Mwangi –C024/401387/2023

Assignment 2: Analyzing HTTP Headers

QUIZ:

a) Use a network packet sniffer (Wireshark or equivalent) to analyze HTTP headers.

b) Capture an HTTP GET and POST request.

c) Identify:

1) Request and response headers.

2) MIME type of the response.

3) HTTP status code and explanation.

Submit a detailed report with screenshots.

Introduction

The following report uses Wireshark to capture and analyse an HTTP GET and POST request to analyse HTTP headers.

Procedure:

1) Tool: Wireshark

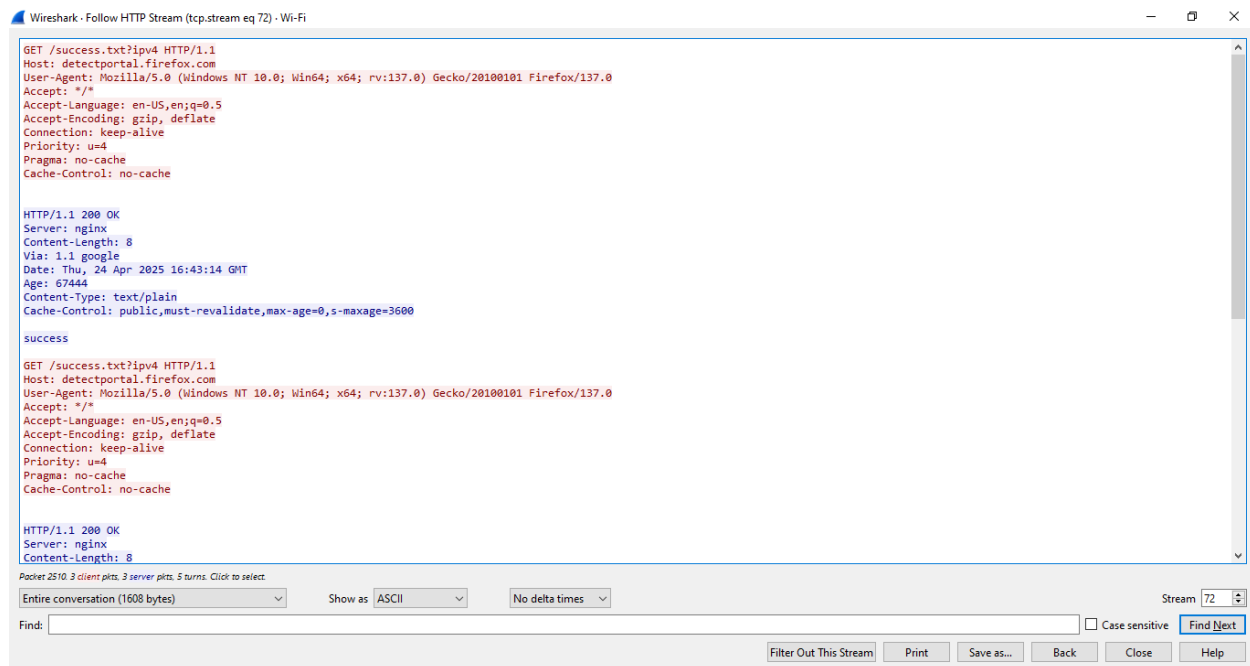
Procedure:

- GET: <http://neverssl.com/>
- POST: Submitted a form on <http://httpbin.org>
- Analyzed packets using Wireshark.

Results and Analysis

a) HTTP GET Request

- Request Headers:



- Response Headers

HTTP/1.1 200 OK

Server: nginx

Content-Length: 8

Via: 1.1 google

Date: Thu, 24 Apr 2025 16:43:14 GMT

Age: 67444

Content-Type: text/plain

Cache-Control: public, must-revalidate, max-age=0, s-maxage=3600

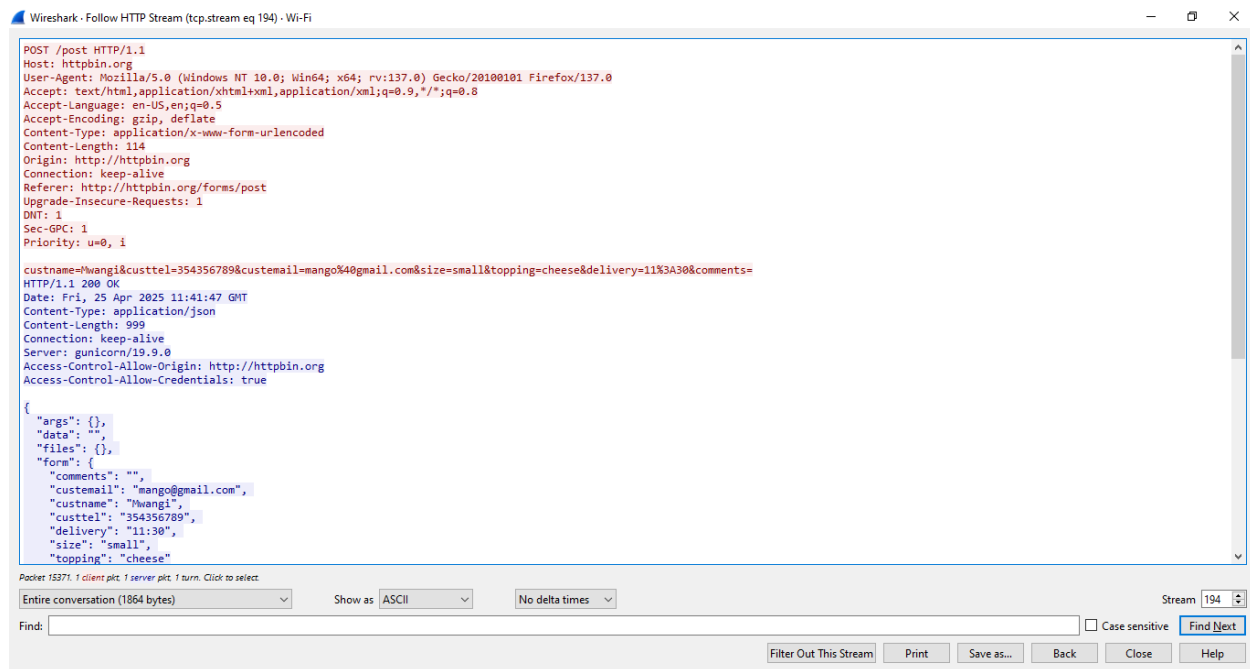
Success

Notes: Gives a HTML page;

MIME Type: text/html—an HTML document .

b) HTTP POST Request

- Request Headers:



Wireshark - Follow HTTP Stream (tcp.stream eq 194) - Wi-Fi

```
POST /post HTTP/1.1
Host: httpbin.org
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:137.0) Gecko/20100101 Firefox/137.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Content-Type: application/x-www-form-urlencoded
Content-Length: 114
Origin: http://httpbin.org
Connection: keep-alive
Referer: http://httpbin.org/forms/post
Upgrade-Insecure-Requests: 1
DNT: 1
Sec-GPC: 1
Priority: u=0, i

custname=Nwagi&custtel=354356789&custemail=mango%40gmail.com&size=small&topping=cheese&delivery=11%3A30&comments=

HTTP/1.1 200 OK
Date: Fri, 25 Apr 2025 11:41:47 GMT
Content-Type: application/json
Content-Length: 999
Connection: keep-alive
Server: gunicorn/19.9.0
Access-Control-Allow-Origin: http://httpbin.org
Access-Control-Allow-Credentials: true

{
  "args": {},
  "data": "",
  "files": {},
  "form": {
    "comments": "",
    "custemail": "mango@gmail.com",
    "custname": "Nwagi",
    "custtel": "354356789",
    "delivery": "11:30",
    "size": "small",
    "topping": "cheese"
  }
}
```

Packet 15371: 1 client pkt, 1 server pkt, 1 turn. Click to select.

Entire conversation (1864 bytes) Show as ASCII No delta times Stream 194

Find: ☐ Case sensitive [Find Next](#)

[Filter Out This Stream](#) [Print](#) [Save as...](#) [Back](#) [Close](#) [Help](#)

Response headers;

HTTP/1.1 200 OK

Date: Fri, 25 Apr 2025 11:41:47 GMT

Content-Type: application/json

Content-Length: 999
Connection: keep-alive
Server: gunicorn/19.9.0
Access-Control-Allow-Origin: <http://httpbin.org>
Access-Control-Allow-Credentials: true

MIME Type: Registration page.

Findings

- Headers: GET retrieves the page while POST submits credentials (name and email).
- MIME Types: Both responses are likely text/html (GET explicitly states it; POST implied).

The analysis successfully showed HTTP GET and POST requests. Wireshark revealed header details, MIME types, and status codes, showing HTTP interactions for retrieving and submitting data